Gengcong Yang

gengcong.y@gmail.com | GitHub | LinkedIn | Google Scholar

Education

University of Texas at Austin (UT Austin) 08/2023 – 12/2024
*Master in **Computer Science*** Austin, United States

Tsinghua University (THU) 08/2019 – 06/2022
*Master in **Control Engineering**, Department of Automation* Beijing, China
Supervisor: Yujiu Yang

National Chiao Tung University (NCTU) 09/2017 – 01/2018
*Exchange Student in **Computer Science*** Hsinchu, Taiwan

Sun Yat-sen University (SYSU) 08/2015 – 06/2019
*Bachelor in **Software Engineering*** Guangzhou, China

Publications

Diamond: Harnessing GPU Resources for Scientific Deep Learning 2025

- Authors: Haotian Xie, Rohan Marwaha, Minu Mathew, Song Bian, **Gengcong Yang**, et al.
- Contributed to the design of a distributed job-submission pipeline enabling scalable, multi-node GPU training on a platform for the scientific machine learning lifecycle; funded by **NSF** and published in eScience.

Probabilistic Modeling of Semantic Ambiguity for Scene Graph Generation 2021

- Authors: **Gengcong Yang**, Jingyi Zhang, Yong Zhang, Baoyuan Wu, Yujiu Yang
- Proposed a probabilistic framework and designed a plug-and-play uncertainty modeling module to capture semantic ambiguity and multiple plausible visual relationships in scene graphs.
- Published in **CVPR 2021**.

Research Experience

Texas Advanced Computing Center, *Graduate Research Assistant* Austin, 01/2024 – 12/2024

- Supervisor: Yinzhi Wang
- Contributed to the early-stage development of Diamond, an NSF-funded, scalable platform supporting the full machine learning research lifecycle on HPC resources.
- Designed and implemented a distributed job-submission pipeline enabling users to containerize environments and train ML models across multiple GPU nodes.

KAUST, *Research Intern* Remote, 02/2021 – 07/2021

- Supervisor: Bernard Ghanem
- Implemented a differentiable RoI Pooling method with CUDA and C++ for the temporal action detection task on videos, achieving an experimental improvement of 6%.

Tencent AI Lab, *Computer Vision Researcher Intern* Shenzhen, 07/2019 – 07/2020

- Supervisor: Baoyuan Wu
- Developed a novel modeling method for polysemous visual-relationship representation and applied for a technology patent.
- Contributed to the design and experimentation of the probabilistic uncertainty modeling framework later published at CVPR 2021.

Industry Experience

TikTok USDS, Emerging Product, *Machine Learning Engineer* — San Jose, 07/2026 – Present

- Coordinated model evaluation for Seedance, a video generation model, ahead of U.S. rollout.
- Supported model serving infrastructure for U.S. deployment, partnering with internal and external cloud providers on provisioning and monitoring.
- Developed agents to automate the team's operational workflows.

Uber, Driver Offer Pricing Team, *Machine Learning Engineer* — Sunnyvale, 01/2025 – 07/2026

- Developed driver fare prediction models used as core pricing signals within Uber's marketplace pricing system.
- Improved observability and reliability infrastructure for pricing prediction pipelines running at scale.

ByteDance, E-commerce, *Machine Learning Engineer* — Shanghai, 06/2022 – 06/2023

- Developed image search models and a retrieval-ranking framework for the TikTok Shop similar-product search project.
- Constructed large-scale datasets for post-pretraining and quantized object detection models to improve serving efficiency.

Tencent, *Machine Learning Engineering Intern* — Beijing, 06/2021 – 08/2021

- Focused on the retrieval stage of the recommendation system for an early-stage short-video platform.
- Built an offline evaluation pipeline for retrieval models and experimented with pairwise training for a two-tower model.

Patents

Visual Relationship Recognition Method Based on Artificial Intelligence

- Inventors: **Gengcong Yang**, Yong Zhang, Baoyuan Wu, Yanbo Fan, Zhifeng Li, Yujiu Yang, Wei Liu
- Patent Number: CN202011108484
- Granted Date: 03/05/2024

Selected Awards

Tsinghua SIGS Professional Practice Scholarship	2022
Tsinghua Excellent Scholarship First Prize	2021
SYSU Outstanding Undergraduate Thesis Award	2019
SYSU Excellent Scholarship Second Prize	2018
National Encouragement Scholarship	2017, 2018
SYSU Xian Weijian Overseas Study Funding	2017
Interdisciplinary Contest In Modeling Honorable Mention	2017
SYSU Excellent Scholarship Third Prize	2016, 2017